

Reinforcement Learning Laboratory

Lab Assignment - 04 - MATLAB Reinforcement Learning



Course Completion Certificate

Aravind Sulakunta

has successfully completed **100%** of the self-paced training course

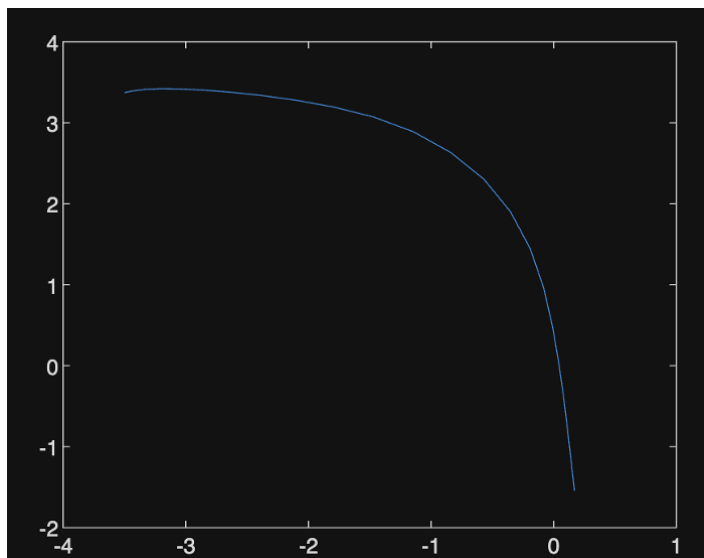
Reinforcement Learning Onramp

A handwritten signature in black ink, reading 'Craig L. Santos'.

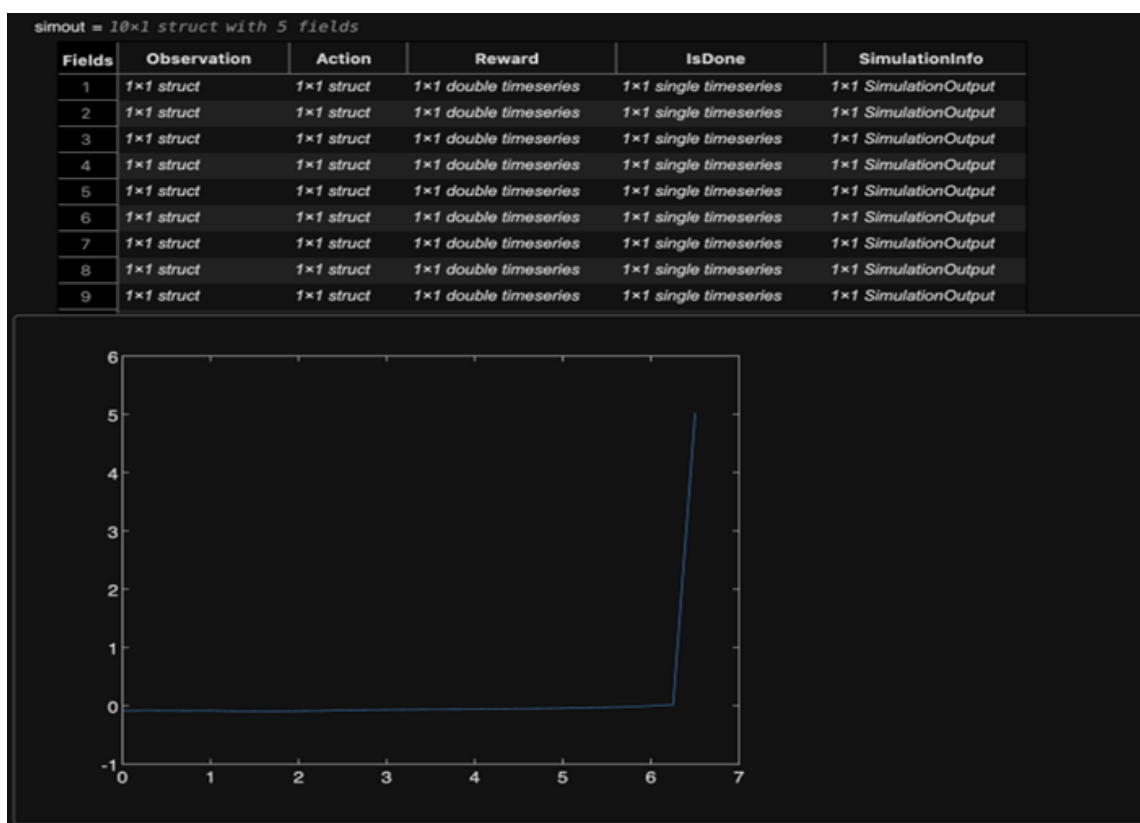
DIRECTOR, TRAINING SERVICES

9 September 2026

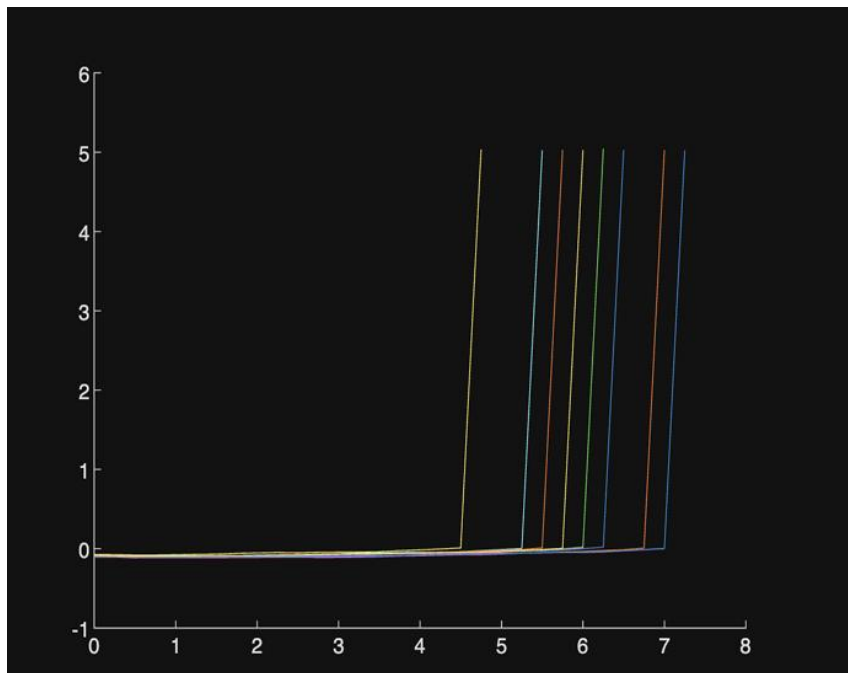
Graphs and Images:



path followed by the agent in the 2D environment during one simulation episode, illustrating how it moves from the starting point toward its destination.



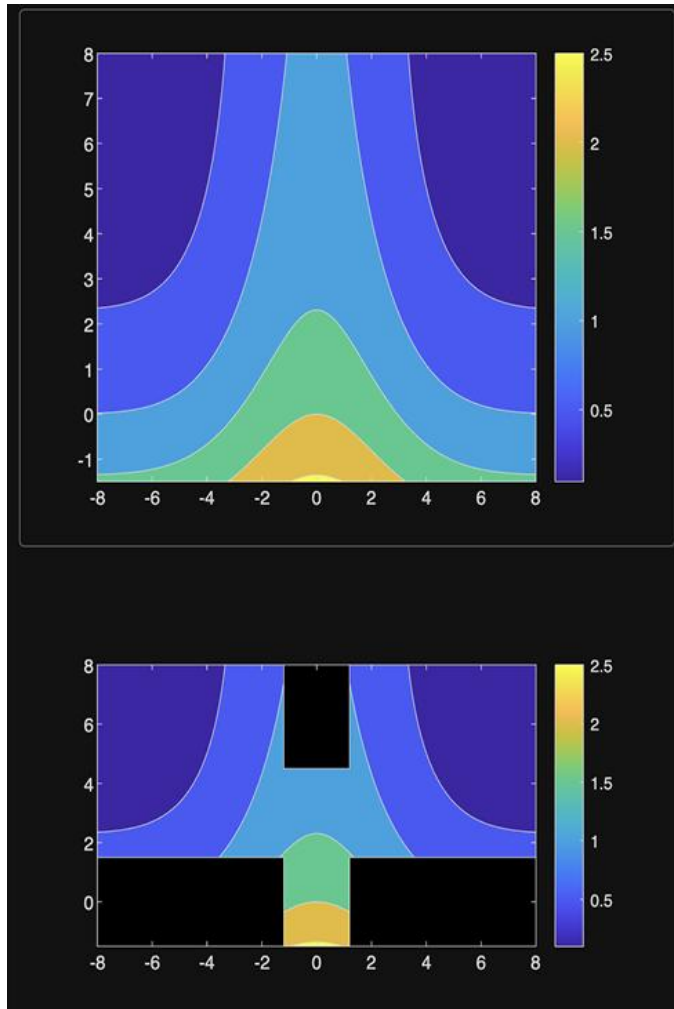
It shows the simulation data (observations, actions, and rewards) along with the agent's movement trajectory during one evaluation episode.



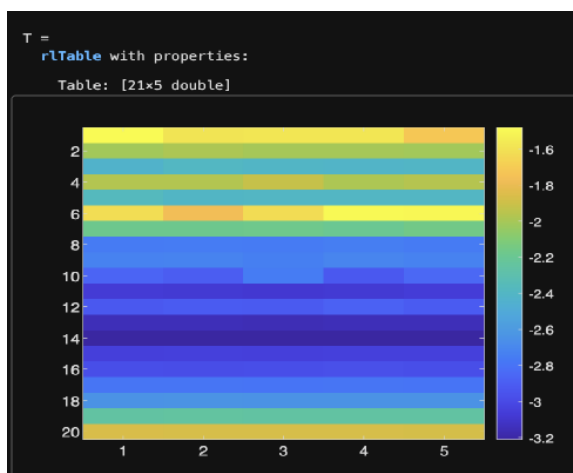
Plots multiple simulated agent trajectories to compare how efficiently the trained policy reaches the goal state across different runs.

Target	Die	Sides						Roll
19	2	2	2	3	3	6	6	2
17	1	1	1	1	5	5	8	8
9	4	2	2	6	6	7	8	2
7	5	3	4	5	7	8	8	4
3	2	2	2	3	3	6	6	6
3	2	2	2	3	3	6	6	2
1	1	1	1	1	5	5	8	1
0	—	—	—	—	—	—	—	—
Target reached in 7 steps								
Target	Die	Sides						Roll
19	2	2	2	3	3	6	6	3
16	4	2	2	6	6	7	8	7
9	4	2	2	6	6	7	8	7
2	2	2	2	3	3	6	6	6
4	3	1	4	4	4	5	7	4
0	—	—	—	—	—	—	—	—
Target reached in 5 steps								

It shows the agent's step-by-step dice rolls and state changes, demonstrating how it reaches the target in a specific number of steps.



It shows the reward/value landscape of the environment, where brighter regions indicate higher value and black areas represent obstacles.



Visualizes a trained agent's Q-table as a heatmap to evaluate the learned action-value function for discrete states and actions.

Observation Table

Parameter	Observation
RL Environment	Continuous control environment – Warehouse Robot Navigation
Agent Used	DDPG (Deep Deterministic Policy Gradient)
Number of Training Episodes	2000 (maximum limit set during training)
Initial Performance	Very low/negative rewards because of frequent collisions and random movement
Final Performance	High and stable rewards, showing efficient navigation to the target
Training Time	Approximately 12 minutes
Training Stopped/Converged At	Episode 485, because the average reward reached the required threshold

20. What type of RL environment is used in the Onramp exercise?

Answer:

A **continuous control environment** is used, where a warehouse robot learns to navigate toward a target.

21. What type of agent is trained?

Answer:

A **DDPG (Deep Deterministic Policy Gradient) agent** is trained.

22. What is the purpose of training the agent?

Answer:

The purpose is to teach the robot an **optimal control policy** so that it can reach the target while avoiding obstacles and collisions.

23. What performance measure is used during training?

Answer:

The main performance measure is the **episode reward**, along with the **moving average reward** over recent episodes.

24. How does the agent's performance change as training progresses?

Answer:

Initially, the agent performs poorly and receives negative rewards because it explores randomly and often collides with obstacles. As training continues, the reward increases and the agent learns a better path toward the target.

Analyze the Learning Curve

Training Stage	Episode Range	Reward/Performance	Observation
Initial	1–50	Highly negative and flat	The agent is mainly exploring. It takes random actions, crashes frequently, and usually fails to reach the target.
Early Training	51–200	Fluctuating with slight improvement	The neural network starts learning. The agent sometimes reaches the target but still has difficulty avoiding obstacles.
Middle Training	201–350	Sharp upward trend	The agent learns a useful policy. Rewards increase quickly as it learns to avoid walls and move toward the target.
Late Training	351–450	Stabilizing near maximum	The curve starts becoming flat, fluctuations reduce, and the moving average reward increases steadily.
Final	451–485	Converged	The average reward reaches the required threshold. The agent consistently navigates successfully, so training stops automatically.

25. What does the learning curve represent?

Answer:

The learning curve shows how the **agent's performance changes during training**. It uses episode rewards and moving average rewards to show how well the agent is learning the task.

26. Why is the reward generally low during the initial training episodes?

Answer:

Because the agent is initially exploring randomly. It makes many mistakes, collides with obstacles, fails to reach the target, and therefore receives negative rewards.

27. What indicates that the agent is learning?

Answer:

A **sharp increase in episode rewards and moving average rewards** indicates that the agent is improving its policy and learning the task.

28. Does the reward increase consistently throughout training? Explain.

Answer:

No. The reward can fluctuate because of random exploration, changes in neural-network weights, and stochastic behavior during training.

29. How can you identify convergence from the learning curve?

Answer:

Convergence can be identified when the learning curve **becomes stable and relatively flat**, with rewards staying near the required maximum/threshold and showing little variation.

30. What could cause fluctuations in the learning curve?

Answer:

Fluctuations can be caused by **high learning rates, large gradient updates, and random exploratory actions**, which may produce sudden increases or decreases in reward.

31. What happens if the agent is trained for a larger number of episodes?

Answer:

The agent can reach a **stable solution**, but training beyond the point of saturation usually gives only small improvements and has diminishing returns.

32. Why is it important to analyze the learning curve rather than considering only the final reward?

Answer:

The learning curve helps us understand the agent's **training stability, learning efficiency, and consistency**. It also shows whether the final high reward comes from proper learning rather than a lucky result.

The DDPG agent starts with poor performance, gradually learns to avoid obstacles and reach the target, and finally converges at Episode 485 with stable performance.